Recurrence Relations, Markov Chains & Geometric Probability Practice Exercises

1 A sequence $\{a_n\}_{n\ge 0}$ satisfies $a_0 = 1$, $a_1 = 3$, and the linear recurrence $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \ge 2$. Compute the exact value of a_5.

a. 31

b. 65

c. 63

d. 59

2 A sequence $\{a_n\}_{n\ge 0}$ satisfies $a_0 = 2$, $a_1 = 4$, and the linear recurrence $a_n = 4a_{n-1} - 3a_{n-2}$ for $n \ge 2$. Compute the exact value of a_5.

a. 82

b. 244

c. 246

d. 238

3 A sequence $\{a_n\}_{n\ge 0}$ satisfies $a_0 = 3$, $a_1 = 6$, and the linear recurrence $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \ge 2$. Compute the exact value of a_5.

a. 96

b. 48

c. 99

d. 90

4 A sequence $\{a_n\}_{n\ge 0}$ satisfies $a_0 = 4$, $a_1 = 7$, and the linear recurrence $a_n = 4a_{n-1} - 3a_{n-2}$ for $n \ge 2$. Compute the exact value of a_5.

a. 124

b. 370

c. 358

d. 367

5 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 1$, $a_1 = 4$, and $a_n = 5a_{n-1} - 6a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 146

b. 500

c. 454

d. 1006

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6 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 2$, $a_1 = 5$, and $a_n = 5a_{n-1} - 6a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 97

b. 275

c. 310

d. 695

7 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 3$, $a_1 = 3$, and $a_n = a_{n-1} + 2a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 63

b. 33

c. 78

d. 3

8 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 1$, $a_1 = 5$, and $a_n = a_{n-1} + 2a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 31

b. 82

c. -3

d. 65

9 A biased coin lands Heads with probability $p = \frac{1}{2}$ and Tails with probability $1 - p = \frac{1}{2}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 4

b. 10

c. 6

d. 7

10 A biased coin lands Heads with probability $p = \frac{1}{3}$ and Tails with probability $1 - p = \frac{2}{3}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 9

b. 12

c. 21

d. 13

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11 A biased coin lands Heads with probability $p = \frac{1}{4}$ and Tails with probability $1 - p = \frac{3}{4}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 20

b. 16

c. 36

d. 21

12 A biased coin lands Heads with probability $p = \frac{1}{5}$ and Tails with probability $1 - p = \frac{4}{5}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 25

b. 55

c. 31

d. 30

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